

CARLETON UNIVERSITY
Department of Systems and Computer Engineering
SYSC 5104. METHODOLOGIES FOR DISCRETE EVENT MODELLING AND SIMULATION

Assignment 2

The goal of this assignment is to show understanding of modeling and simulation using the Cell-DEVS formalism, and application of the techniques using the Cadmium tool. You must identify a real system that can be represented using Cell-DEVS, build a cellular model of the system and run simulations of the system under analysis. The real system under analysis can be any natural or artificial system.

Part I:

The first stage in the assignment will be to identify one of such systems in the Cell Spaces modeling literature. The first stage is to select a model of choice from the list of models in the course, or to identify a real system of choice that can be described as a Cell-DEVS model to developed from scratch. If you build your own model, you can use any of the available resources (libraries, the Internet, digital libraries, etc.), and look for examples of different cellular models. You should provide a copy of the chosen material, and provide a one-page conceptual model description, including a brief sketch of the model structure if a coupled Cell-DEVS is chosen. If you choose your own model, you should provide the article used as a basis for the model.

The model should be able to be represented as a Cell-DEVS model, to be later implemented in Cadmium. Students working in teams must define more complex models (maximum: three members per team).

Part II

After this stage is finished, the conceptual model will be used as a basis to develop the final work. You should:

- . Write the formal specifications for the Cell-DEVS models (and for the DEVS models, if any).
- . Using the specification as a basis, build the models using Cadmium
- . Once the models have been tested, build an experimental frame, and execute various simulation examples, showing the reaction of the model to different inputs than those defined in the specifications.
- . Provide a graphical output using the Cell-DEVS Web Viewer (<https://devssim.carleton.ca/cell-devs-viewer/>)
- . Generate a video of interesting results using the Web Viewer.

Write a report showing these execution results and analyzing the behavior of the model according to the specifications.

Important dates:

March 11. Assignment 2 requirement document delivery (earlier submissions are encouraged).

March 24. **Assignment 2 Report delivery.**

Deliverables and grades

Part I

This first activity represents 15% of the final mark. Those students not able to provide a model by themselves, will be given one with the proposed characteristics (but they lose the opportunity to work on a model of their choice). If the real system you want to analyze is too complex, select a subset of the real system, and work with it. The deliverable is the one-page report associated to this stage of the assignment, including the corresponding reference (Word or PDF format). Those students picking their own model from the literature will get a 20% bonus.

Part II

This specification document is worth 25% of the final mark. It will include the model used in the third stage.

The simulation results and the implementation models are worth 60% of the final mark. You should include a final report explaining the execution results of your model, and any variations you have done to the original specification, showing the new results obtained. It should include a brief description of each of the models (as a conceptual model definition), besides the formal specification.

- Use meaningful and self-descriptive names for every file included. Consider using names that represent the models you built (DO NOT use “Assign1.doc”, or “assign1.h”; neither “MyNameAssign1.cpp”, etc... Use a name that represents the content of the model).

DEVModelsForm

You should fill out this form and submit it with your report.

Your should prepare only one final report, which will be organized as a mix of the documents you created in Parts I-III, all combined in a single document. This document will include:

- Part I

- Part II, which should explain the execution results of your model, any variations you have done to the original specification, and the new results obtained by varying the original experimental framework.

Remember that the documents for the 2 parts should be combined in a single document to be delivered, included in the same repository with the simulation software and documentation.

Besides the report, you should include the software developed. You should include:

- Source code
- **Scripts** to execute each of the models using different experimental frameworks
- Any 'read.me' files explaining the detailed behavior for each one of the scripts provided.
- Log files showing the execution results
- Your final **report** document (in .doc or .rtf format)
- The **DEVModelsForm** describing the model and giving author information

All of this information should be included in a GitHub repository. Your repository will be cloned; include instructions to clone and run the model if needed.

Use meaningful and self-descriptive names for every file included. Consider using names that represent the models you built (DO NOT use “Assign1.doc”, “assign1.hpp”, “MyNameAssign1.cpp”, etc... Use a name that represents the content of the model).

The reports will be analyzed as follows:

- . The contents of model will be executed
- . The video files submitted will be viewed
- . The results will be visualized using the Web Viewer.
- . The examples will be run using the scripts you provided.
- . After these basic steps are carried out, different changes will be studied, in order to find possible errors.

In order to ensure proper execution, try to execute these same steps by yourselves, before delivering the models.

Include detailed information on each one of the scripts provided.

Check the Life Game model in the Examples webpage to see what is expected.

CHECKLIST

Read carefully the requirements specified before, and to everything explained there; if something is missing, the assignment will be returned unread.

Tick each one of these items, and verify you comply with them.

- ___ You have combined the 2 parts of the report in ONE report. Each part has a meaningful name (not Part I, II, but a descriptive title explaining the content of the subchapter).
- ___ You have included the report document in your zip file
- ___ You have **only** included the needed files:
 - ___ Source code (*.h/*.cpp) for each of the atomic models (if you wrote a DEVS model as part of yours)
 - ___ Scripts to execute each of the models using different experimental frameworks
 - ___ Explanations for the detailed behavior for each one of the scripts provided.
 - ___ .log files showing the execution results
 - ___ Your final report document (.doc files, or .pdf versions)
 - ___ Video files of visualization results
- ___ You have put meaningful and self-descriptive names to each of the files (including the zip file and folders used)
- ___ You have imported your zip in a “clean” directory and tried to recompile and run each of the tests, and you were successful in each of them.
- ___ You have checked the “Life” model to see an example of what is required.
- ___ You have filled out and submitted the DevsModelForm file.